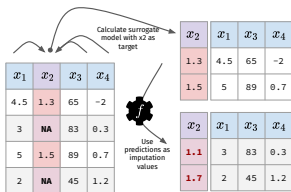




Applied Machine Learning

Imputation: Advanced Methods and Pitfalls

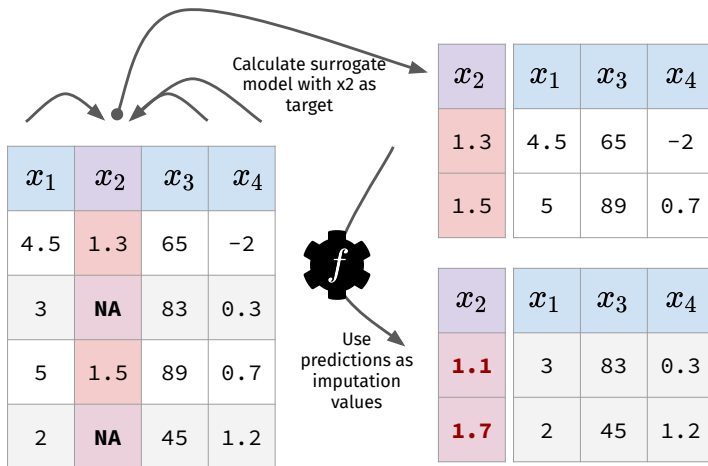


Learning goals

- Based imputation strategies
- Practical considerations for missing values

MODEL-BASED IMPUTATION

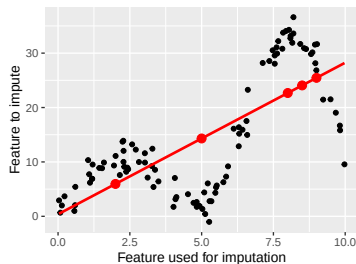
Instead of imputing a single value or sampling values it is desirable to take advantage of structure and correlation between features.



MODEL-BASED IMPUTATION: DRAWBACKS

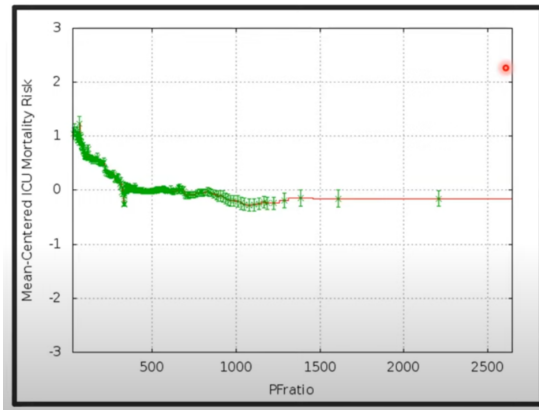


- Choice of surrogate model has high influence on the imputation:



- Surrogate model should handle missing values itself, otherwise imputation *loop* may be necessary.
- Surrogate model hyperparameters can be tuned and can be different for each feature to impute.

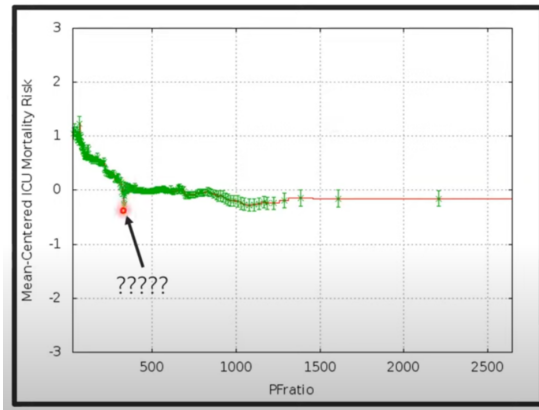
FAILED IMPUTATION: ICU OXYGENATION (I)



Source: ▶ [Rich Caruana's talk at AutoML Conference 2024](#) (example at 43'31)

- ICU mortality vs. **PF ratio** (oxygenation)
- ≈ 0 : no breathing; ~ 1000 : healthy
- Expect *decreasing* risk with PF

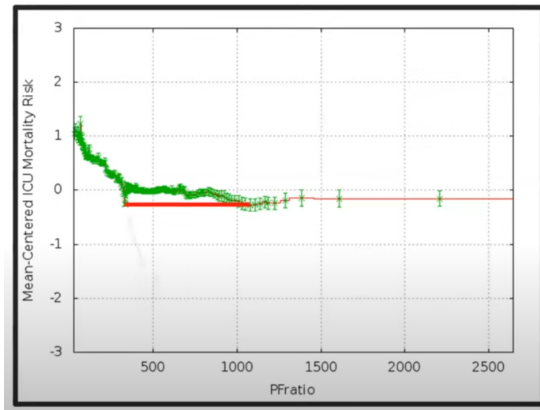
FAILED IMPUTATION: ICU OXYGENATION (II)



Source: ▶ [Rich Caruana's talk at AutoML Conference 2024](#) (example at 43'31)

- Local dip in risk at one PF value
- Contradicts physiology and trend
- **Question:** Why so low?

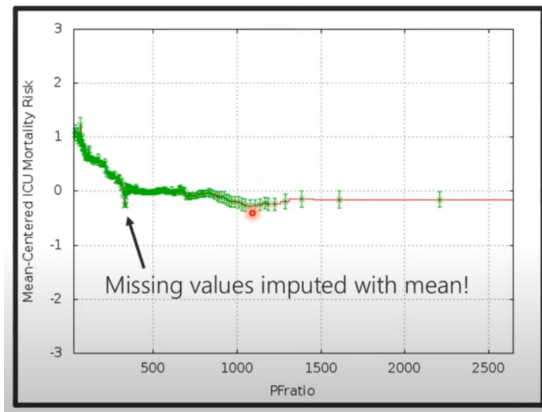
FAILED IMPUTATION: ICU OXYGENATION (III)



Source: ▶ [Rich Caruana's talk at AutoML Conference 2024](#) (example at 43'31)

- Missing PF values imputed by the mean (collapse to **one** PF point)
- Model learns spuriously *low* risk there
- Imputation ignores MNAR/MAR structure

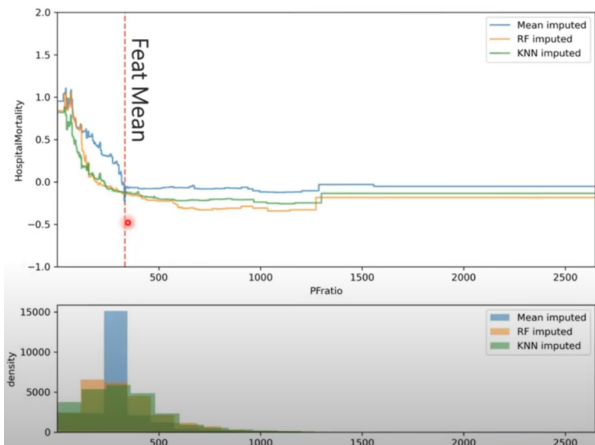
FAILED IMPUTATION: ICU OXYGENATION (IV)



Source: ▶ *Rich Caruana's talk at AutoML Conference 2024* (example at 43'31)

- With imputation: $\sim 60\%$ healthy mapped to $PF < 1000$
- Label-feature mismatch near imputed PF
- Risk landscape shifts toward “healthier”
- **Should we use a more complex imputer?**

FAILED IMPUTATION: ICU OXYGENATION (V)



Source: ▶ [Rich Caruana's talk at AutoML Conference 2024](#) (example at 43'31)

- “Smarter” imputers *spread* healthy into $PF < 1000$
- Even worse: Final model underestimates risk for the sick
- Prefer **no fill** + missingness indicator
- If imputing: fit/tune *inside* CV; consider MNAR

TESTS FOR MISSINGNESS MECHANISMS



MAR vs MCAR

- Little's test
- Logistic regression for missingness
 - 1 For each column, create a new classification task: predict if the value is missing or not
 - 2 Use the remaining columns as features
 - 3 Fit logistic regression
 - 4 Examine coefficients and p-values

MNAR: no test available

CENSORING MECHANISM

- **Left censoring:** True value below a threshold; exact value unknown.
- **Interval censoring:** True value lies within a known interval.
- **Right censoring:** True value above a threshold; exact value unknown.



CENSORING MECHANISM



- **Left censoring:** True value below a threshold; exact value unknown.
- **Interval censoring:** True value lies within a known interval.
- **Right censoring:** True value above a threshold; exact value unknown.
- **Type I censoring:** Experiment ends at pre-specified time; remaining subjects are right-censored.
- **Type II censoring:** Experiment ends after a fixed number of failures; remaining subjects are right-censored.
- **Random (non-informative) censoring:** Censoring times independent of failure times; observe minimum of failure and censoring time.

TAKEAWAYS OF A RECENT BENCHMARK



Imputation for prediction: beware of diminishing return. Le Morvan and Varoquaux, ICLR 2025 [► Morvan and Varoquaux 2024](#)

- Better imputation performance does not result in better prediction performance for MNAR (w/o missingness indicator)
- Better imputation might be irrelevant b/c information is available in other features, unimportant, even with imputation, it might be hard to learn a good downstream model
- More expressive models benefit less from imputation
- Best on average: missForest + XGBoost + masking
- Open Questions:
 - performance of random draws
 - performance of multiple imputation
 - impact of performance on explainability and calibration
 - performance of models that can learn directly with missing value (advanced transformer architectures)

TAKEAWAYS



- There exist different reasons for missing data
 - There exist different missingness mechanisms
 - Different downstream tasks ask for different imputation strategies
 - Different data modalities ask for different imputation strategies
 - Understanding the data generating process helps deciding on the imputation strategy
 - Common imputation strategies:
 - Constant value: mean, median
 - Imputation by sampling
 - Missingness indicator
 - Model-based
- imputation selection is a CASH problem